18/12/24 M/S

USN | | | | | | | | | | [22CSEDE11]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

BACKEND PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A	**5 x2=10**		
1.	**a.**	Explain the tasks and responsibilities of Back End Developer.	L2	CO1	**02**
	b.	List is the advantages of using NPM.	L1	CO2	**02**
	c.	Explain the installation process of Express.	L2	CO3	**02**
	d.	List the use of learning Reactjs in detail.	L2	CO4	**02**
	e.	Explain about CRUD operations.	L2	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the difference between Front End and Back End Developer.	L2	CO1	**05**
		OR			
	b.	Explain the tools and Technologies of Back-end Development.	L2	CO1	**05**
3.	**a.**	Explain in detail REPL and its commands.	L2	CO2	**05**
		OR			
	b.	List the key features of Nodejs.	L1	CO2	**05**
4.	**a.**	Explain with example code the role of Routing and Middleware in Expressjs.	L2	CO3	**05**
		OR			
	b.	Write about Restful routing with Express.	L1	CO3	**05**
5.	**a.**	Explain when and how to use react components.	L2	CO4	**05**
		OR			
	b.	Differentiate between Props and States in Reactjs.	L2	CO4	**05**
6.	**a.**	Explain Mongoose DB with its functionality.	L2	CO5	**05**
		OR			
	b.	Explain the concept of data serialization.	L2	CO5	**05**
7.	**a.**	Explain the difference of Routing in Nodejs and Expressjs.	L2	CO3	**05**
		OR			
	b.	Explain about MongoDB connections and commands.	L2	CO6	**05**

Deemed-to-be University Learning Resource Centre Jain Global Campus Ramanagara

		PART-C	**6 X10=60**		
8.	**a.**	Explain the tools and Technologies which require to become a full stack developer.	L2	CO1	**10**
		OR			
	b.	Explain the role of Servers, Databases and API in Backend Developers.	L2	CO1	**10**
9.	**a.**	Explain about the Non-blocking and Asynchronous concept and how it is useful in Nodejs Architecture.	L2	CO2	**10**
		OR			
	b.	Explain in brief installation process Nodejs and write a code to run Nodejs application.	L2	CO2	**10**
10.	**a.**	Create an API and filter depending upon the user requirement.	L5	CO3	**10**
		OR			
	b.	Illustrate a code to create a server with Express.	L4	CO3	**10**
11.	**a.**	Explain about hooks and explain the most commonly used hooks	L2	CO4	**10**
		OR			
	b.	Contrast between Props and States.	L2	CO4	**10**
12.	**a.**	Explain about NoSQL database and its importance	L2	CO5	**10**
		OR			
	b.	Explain Conceptual techniques for NoSQL data modeling.	L2	CO5	**10**
13.	**a.**	Illustrate a code to display the html pages using Express.	L4	CO3	**10**
		OR			
	b.	Explain about Mongoose installation and Schema creation.	L2	CO6	**10**